MATH REVIEW/BOOTCAMP - Project 0

CURIE SP26 - Programming in Computational Quantum Chemistry

1 Some Resources

For linear algebra I can't recommend watching the video series by 3Blue1Brown enough. If the playlist is overwhelming, here are the videos I highly recommend watching:

- Matrix and Vector Operations, Linear Transformations (Ch 1 - Ch 4)
- Change of Basis (Ch 13)
- Eigenvalues, Eigenvectors (Ch 14)
- Abstraction/Vector Spaces (Ch 16)

If you have additional time these are nice to watch as well:

- Inverse matricies, rank, column space, null space (Ch 7)
- Determinants (Ch 6) (*we will use determinants for another property other than areas/volumes*)
- Dot Products (Ch 9) (*This video has way more detail than what we need*)

For vector calculus and multivariable calculus here are some solid online notes and a good playlist and lecture series.

2 Differentiation and Integration

2.1 Differentiation

Problem 2.1(a) The function $f(x, y) = x^2y^2$ describes a parabola analogous to $f(x) = x^2$. Find the analytical derivatives in respect to both x and y, denoted $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

Problem 2.1(b) The function $f(x, y) = x^2 - y^2$ describes a saddle function. Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

Problem 2.1(c) Evaluate the gradient of $f(x, y)$ in both problems 2.1(a) and 2.1(b). Plot the gradient vector on the contour plot over several (x, y) values and observe that the vector is orthogonal to the contour lines, pointing in the direction of steepest descent. Refer to section 5.1.

Problem 2.1(d) The Lennard-Jones potential is a simple potential for describing London dispersion forces given by the formula:

$$E(r) = 4\varepsilon \left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6} \right]$$
$$r = \sqrt{x^2 + y^2 + z^2}$$

Evaluate the analytical gradient of the Lennard-Jones potential. You answer should be a gradient vector of dimensions 3×1. (Be mindful of the chain rule!) From physics, we know that the derivative of the energy corresponds to the $\nabla U(x, y, z) = -F(x, y, z)$. By implementing this formula you can (and will in Project 1!) optimize the geometry of a cluster of atoms by update the positions of the atoms through iterating over a step size.

2.2 Integration

Problem 2.2(a) The normalized wavefunction for the ground state hydrogen atom is:

$$\psi_{1s}(r, \theta, \varphi) = \frac{1}{\sqrt{\pi}\, a_0^{3/2}} e^{-r/a_0}$$

Show that this function is normalized by showing that the normalization constant computes to 1:

$$N = \int_{\mathbb{R}^3} |\psi_{1s}|^2 dV = 1$$

Compute the integral over all space given by the bounds $r : [0, \infty),\ \theta : [0, 2\pi],\ \varphi : [0, \pi]$. The following integral will be useful:

$$\int_0^\infty x^n e^{-\alpha x} dx = \frac{\Gamma(n+1)}{\alpha^{n+1}} \qquad \Gamma(n) = (n-1)!$$

Where $\Gamma(n)$ is a gamma functinon. Refer to section 5.2 for notes on integration and 5.3 for normalization of functions.

3 Linear Algebra

3.1 Matrix and Vector Operations

Problem 3.1 (a) Express the following system of linear equations:

$$\begin{aligned} x_1 + 3x_2 + 2x_3 &= b_1 \\ 2x_1 + 2x_2 + 2x_3 &= b_2 \\ 3x_1 + 5x_2 + 6x_3 &= b_3 \end{aligned}$$

as a matrix-vector multiplication $A\mathbf{x} = \mathbf{b}$ and compute vector $\mathbf{b}$ if

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{x}_2 = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$$

Check your answer by solving the equation in python. (*Note the difference in the* $*$ *and* @ *operations.*)

Problem 3.1 (b) Evaluate $C = AB$ given the matrices A and B:

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & 5 \\ -2 & 1 & 1 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 2 & 0 \\ -3 & 1 & 4 \\ 2 & -1 & 3 \end{bmatrix}$$

Check your answer by doing the matrix-matrix multiplication in python.

Problem 3.1 (c) Find the inverse of matrix A and the transpose of matrix B given the matrices A and B:

$$A = \begin{bmatrix} 3 & -2 \\ 5 & 1 \end{bmatrix} \qquad B = \begin{bmatrix} 4 & -1 \\ 0 & 7 \\ -3 & 2 \end{bmatrix}$$

3.2 Norms

Problem 3.2(a) Find the norm and normalize vector $\mathbf{v}$. Compute it again using python. For matrix $\mathbf{A}$, use the np.linalg.norm() function in Python to compute the Frobenius norm and normalize the matrix.

$$\mathbf{x} = \begin{bmatrix} 3 \\ -1 \\ 2 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{A} = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$$

Problem 3.2(b) Normalize the functions $\psi_1(r,\theta,\phi) = e^{-Zr}$ and $\psi_2(r,\theta,\phi) = re^{-Zr}$. Also check if these basis functions are orthogonal. Refer to 2.2 (a) for integral formulas and sections 5.2 and 5.3.

3.3 Eigenvalues, Eigenvectors

Problem 3.3(a) Compute the eigenvalues and eigenvectors of the matrix in Problem 2.1 (a). Confirm your answer by using np.linalg.eig().

Problem 3.3(b) Show that the operator

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\hat{d}^2}{dx^2} \tag{1}$$

has eigenvalues and eigenfunctions:

$$E_n = \frac{n^2h^2}{8mL^2} \qquad \psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)$$

where the reduced Planck constant $\hbar^2 = h/2\pi$. Show likewise for the following operator

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\hat{d}^2}{dx^2} + \frac{1}{2}m\omega^2\hat{x}^2 \tag{2}$$

with eigenvalue and eigenfunctions:

$$E = \hbar\omega\left(\frac{1}{2}+n\right) \qquad \psi_n(x) = \frac{1}{\sqrt{2^n n!}}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega x^2}{2\hbar}} H_n\left(\sqrt{\frac{m\omega}{\hbar}}\,x\right)$$

for $n = 0$. I recommend substituting $\alpha = m\omega/\hbar$ to simplify the problem. Note that the Hermite Polynomial $H_n(x)$ for $n = 0$ is: $H_0(x) = 1$.

These are solution for the Schrödinger Equation $\hat{H}\psi = E\psi$ for the Particle in a Box (1) and the Harmonic Oscillator (2). We will explore Schrödinger Equation and these systems in more detail in Project 2.

4 Differential Equations

Problem 4(a) Find the general solution to the differential equation (*2nd-order linear ODE*)

$$\frac{dy^2}{dx^2} + \lambda y^2 = 0$$

And find the conditions for the integration constants A and B and λ that satisfies the boundary condition $y(0) = y(1) = 0$ (*This is called a Dirichlet Boundary Condition*)

Problem 4(b) Solve the system of ODEs with initial conditions

$$\begin{cases} \dfrac{d}{dt}x_1(t) = 2\,x_2(t) \\ \dfrac{d}{dt}x_2(t) = 4\,x_1(t) \end{cases} \qquad \begin{cases} x_1(0) = 1 \\ x_2(0) = 3 \end{cases}$$

As a decoupled Eigenvalue problem:

$$\mathbf{x}'(t) = \begin{bmatrix} 0 & 2 \\ 4 & 0 \end{bmatrix} \mathbf{x}(t) \qquad \mathbf{x}(0) = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

where

$$\mathbf{x}'(t) = \begin{bmatrix} \frac{d}{dt}\,x_1(t) \\ \frac{d}{dt}\,x_2(t) \end{bmatrix} \qquad \mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

5 Additional Notes

5.1 Gradients, Directional Derivatives

A gradient is a vector whose components are partial derivatives. The gradient can be extended beyond three variables to a function of n variables. Evaluated at some initial point $(x_1, x_2, ...x_n)$, the gradient returns a vector pointed at the direction of steepest descent.

$$\nabla = \begin{bmatrix} \frac{\partial}{\partial x_1} \\ \frac{\partial}{\partial x_2} \\ \vdots \\ \frac{\partial}{\partial x_0} \end{bmatrix} \qquad \nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

In dimensions greater than two, derivatives can be evaluated in a direction. This directional derivative $D_{\mathbf{v}} f$ can be computed by taking the dot product of the gradient with some unit vector $D_v f = \nabla f \cdot v$, where v is a unit vector pointed in a desired direction.

5.2 Multivariable Integration

Computing integrals over multiple variables is conceptually identical to differentiation over multiple variables. For reasons, however, an additional factor is required when integrating over different coordinates.

$$\text{Polar Coordinates: } F(r\theta) = \iint f(r,\theta) \quad \underline{r} \quad dr\, d\theta$$

$$\text{Cylindrical Coordinates: } F(r,\theta,z) = \iiint f(r,\theta,z) \quad \underline{r} \quad dr\, d\theta\, dz$$

$$\text{Spherical Coordinates: } F(r,\theta,\varphi) = \iiint f(r,\theta,\varphi) \quad \underline{r^2 \sin\varphi} \quad dr\, d\theta\, d\varphi$$

5.3 Norms, Inner Products, Complex Conjugate

You may already be familiar with finding the norm of a vector by taking the dot product $\|\mathbf{v}\| = \mathbf{v} \cdot \mathbf{v} = \mathbf{v}^T \mathbf{v}$ and normalizing it by dividing the vector by its norm $\frac{\mathbf{v}}{\|\mathbf{v}\|}$. Normalization can be defined in many ways but for us, we will define the norm of some function ψ as the inner product:

$$N = \langle \psi | \psi \rangle = \int_{-\infty}^{\infty} |\psi|^2 \, dV = \int_{-\infty}^{\infty} \psi^* \psi \, dV$$

An inner product is a generalization of the dot product for any vectors (like functions). ψ^* denotes the complex conjugate of ψ, where all complex numbers are multiplied by -1. You will also see †, which applies a complex conjugate and a transpose for vectors and matricies. For example:

$$\psi(x,t) = e^{-(x-ct)^2+ik(x-ct)}$$
$$\psi^*(x,t) = e^{-(x-ct)^2-ik(x-ct)}$$

$$|\psi\rangle = \begin{bmatrix} -i \\ 4 \\ 3i \end{bmatrix} \begin{matrix} |\varphi_1\rangle \\ |\varphi_2\rangle \\ |\varphi_3\rangle \end{matrix} = -i|\varphi_1\rangle + 4|\varphi_2\rangle + 3i|\varphi_3\rangle$$
$$\langle\psi| = (|\psi\rangle)^\dagger = \begin{bmatrix} i & 4 & -3i \end{bmatrix} = i\langle\varphi_1| + 4\langle\varphi_2| - 3i\langle\varphi_3|$$

Once you obtain the norm, a function can be normalized similarly to vectors by multiplying $1/N$. (*Note: the vector $\langle\psi|$ is a dual vector of $|\psi\rangle$ for both real and complex values*)

5.4 Symmetric and Hermitian matrices

A matrix is symmetric if $A^T = A$. For example:

$$A = \begin{pmatrix} \varepsilon & t \\ t & \varepsilon \end{pmatrix} \qquad B = \begin{pmatrix} 2 & 4 & 1 \\ 4 & -1 & 0 \\ 1 & 0 & 3 \end{pmatrix}$$

Furthermore, a matrix is Hermitian if $A^\dagger = A$. For example:

$$A = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$$

If a real matrix is symmetric, it is also Hermitian. Hermitian matrices are important in quantum mechanics due to their neat properties stated by something called the spectral theorem. If a matrix is Hermitian, (1) it has real eigenvalues, (2) it has an orthonormal eigenbasis, and (3) it can be diagonalized with a unitary matrix. All observables in quantum mecahnics are represented by Hermitian operators. For now, (1) and (2) will be most applicable for us because (1) we don't want imaginary Energy/observables and (2) we want to be able to describe any state with the complete set of eigenvectors.